

LogPilot - AI-Powered Log Analysis CLI

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Description

A pip-installable CLI for intelligent log analysis. Parses logs, detects anomalies using statistical methods and clustering, identifies root causes, generates incident reports.

Tech Stack

click, scikit-learn, pandas, numpy, pydantic, rich, pytest

Features

- Multi-format log parsing
- Statistical anomaly detection
- Pattern clustering
- Root cause analysis
- Rich terminal output
- Incident report generation

Sample Input

Log files to analyze

```
logpilot analyze --input app.log --format auto
```

Sample Output

Anomaly report with root cause analysis

```
logpilot report --input app.log --output incident.md
```

Project Files

```
.gitignore
LICENSE
project-idea.json
pyproject.toml
README.md
.github/workflows/test.yml
src/logpilot/cli.py
src/logpilot/correlator.py
src/logpilot/detector.py
src/logpilot/display.py
```

```
src/logpilot/generator.py
src/logpilot/models.py
src/logpilot/parser.py
src/logpilot/py.typed
src/logpilot/reporter.py
src/logpilot/__init__.py
tests/conftest.py
tests/test_cli.py
tests/test_correlator.py
tests/test_detector.py
tests/test_generator.py
tests/test_parser.py
tests/test_reporter.py
tests/__init__.py
```

Test Results

PASSED

Run: 34 | Passed: 34 | Failed: 0

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